

Tutorial 1: Student Feedback Application

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Course Name: 24CSC42 - Full Stack Development

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1. Scenario

The objective of this tutorial is to develop a responsive 'Student Feedback Application' using React.js. The application features a split-pane layout: a feedback submission form on the left and a dynamic list of received responses on the right. Key features include real-time form validation (email format and 10-digit mobile numbers), local state management using React Hooks (useState), and a professional UI styled with a Navy and Gold theme.

2. VS Code Project Structure

```
student-feedback-app/  
├── public/  
│   └── index.html  
├── src/  
│   ├── components/  
│   │   ├── FeedbackForm.js  
│   │   └── FeedbackList.js  
│   ├── App.js  
│   ├── index.css  
│   └── index.js
```

```
├─ package.json
└─ README.md
```

3. Source Code

App.js

```
import React, { useState } from 'react';
import FeedbackForm from './components/FeedbackForm';
import FeedbackList from './components/FeedbackList';

function App() {
  const [feedbacks, setFeedbacks] = useState([]);

  const handleFeedbackSubmit = (newFeedback) => {
    setFeedbacks((prev) => [newFeedback, ...prev]);
  };

  return (
    <div>
      <header className="app-header">
        <div className="college-logo">KEC</div>
        <div className="header-text">
          <div className="college-name">Kongu Engineering College,
Erode - 638060</div>
          <div className="header-title">Student Feedback
Application</div>
          <div className="header-sub">24CSC42 Full Stack Development |
Tutorial 1</div>
        </div>
      </header>
      <main className="main-content">
        <FeedbackForm onSubmit={handleFeedbackSubmit} />
        <FeedbackList feedbacks={feedbacks} />
      </main>
    </div>
  );
}
```

```
export default App;
```

FeedbackForm.js

```
import React, { useState } from 'react';

const EMPTY_FORM = { name: '', email: '', contact: '', message: '' };

function FeedbackForm({ onSubmit }) {
  const [form, setForm] = useState(EMPTY_FORM);
  const [errors, setErrors] = useState({});
  const [submitted, setSubmitted] = useState(false);

  const validate = () => {
    const newErrors = {};
    if (!form.name.trim()) newErrors.name = 'Student name is required.';
    if (!/\S+@\S+\.\S+/.test(form.email)) newErrors.email = 'Valid email required.';
    if (!/^d{10}$/.test(form.contact)) newErrors.contact = '10-digit number required.';
    if (!form.message.trim()) newErrors.message = 'Feedback message cannot be empty.';
    return newErrors;
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    const validationErrors = validate();
    if (Object.keys(validationErrors).length > 0) {
      setErrors(validationErrors);
      return;
    }
    onSubmit({ id: Date.now(), ...form, timestamp: new Date().toLocaleString() });
    setForm(EMPTY_FORM);
    setSubmitted(true);
    setTimeout(() => setSubmitted(false), 3000);
  };
}
```

```

};

return (
  <div className="form-card">
    { /* Form Implementation... */ }
  </div>
);
}

```

4. Output Screenshots

4. Final Feedback List

The screenshot shows a web application titled "4. Final Feedback List". The header is dark blue with "KEC" on the left and "Kongu Engineering College, Erode Student Feedback Application" on the right. The main content area has a light orange background and contains two white cards with orange borders. The left card, titled "Submit Feedback", has a text input field labeled "Student Name". The right card, titled "Responses Received", displays a feedback entry from "John Doe" with the text "Great course! The UI is very intuitive."

3. Success Toast Message

KEC

Kongu Engineering College, Erode
Student Feedback Application

Submit Feedback

Student Name

Email ID

Responses Received

2. Form Validation Errors

KEC

Kongu Engineering College, Erode
Student Feedback Application

Submit Feedback

Student Name

Student name is required.

Email ID

wrong-email

Enter a valid email address.

Contact No

123

Enter a valid 10-digit number.

Responses Received

1. Initial Application View

KEC

Kongu Engineering College, Erode
Student Feedback Application

Submit Feedback

Student Name

Email ID

Contact No

Responses Received

No feedback submitted yet.